

CDCL Hardness Prediction on Standard SAT Families

A Benchmark for Graph-Local and Defect-Field Features

Christof Krieg

MAAT Project – Independent Research

<https://maat-research.com>

2026

Abstract

We benchmark graph-local and multi-scale defect-field features against CDCL solver hardness on 395 deterministically generated SAT instances from six standard families: random 3-SAT, planted 3-SAT, modular 3-SAT, 3-XOR CNF, graph coloring, and pigeonhole formulas. Hardness is measured from PySAT implementations of Glucose3 and MiniSat22 using the predeclared target

$$\log(1 + \overline{C} + 0.10 \overline{D} + 0.01 \overline{P}),$$

where $\overline{C}$, $\overline{D}$, and $\overline{P}$ are mean conflicts, decisions, and propagations across the two CDCL solvers. The comparison includes density-only, standard graph, scalar structural, graph-local structural, multi-scale defect-field, and shuffled-defect-null feature sets.

The strongest random-forest baseline is the standard graph feature set ($R^2 = 0.2742$, Spearman $\rho = 0.4254$). The proposed multi-scale defect-field representation is close but does not dominate ($R^2 = 0.2621$, $\rho = 0.3984$), while graph-local structural features reach $R^2 = 0.2573$ and $\rho = 0.4170$. Scalar structural compression is weaker ($R^2 = 0.1062$), density-only fails ($R^2 = -0.0894$), and shuffled defect fields collapse ($R^2 = -0.0609$, $\rho \simeq -0.007$).

The decisive negative result is leave-family-out transfer: all feature sets fail to extrapolate cleanly across canonical SAT families. Thus the proposed defect-field features contain real CDCL-hardness signal beyond shuffled nulls and scalar compression, but they remain close to classical graph geometry and do not yet define a family-independent hardness theory.

1 Purpose

Papers 50 and 50b identified a weakness in the earlier SAT programme: global scalar compression did not capture SAT hardness well. Those papers also used a deterministic DPLL node-count target. That choice was transparent and reproducible, but it is not representative of modern SAT solving, which is dominated by CDCL solvers with learned clauses, restarts, and conflict analysis.

The purpose of Paper 55 is therefore narrow:

test whether graph-local and multi-scale defect-field features retain hardness signal when the target is replaced by CDCL solver statistics.

This is not a proof-complexity paper, not a SAT Competition result, and not a claim about industrial SAT. It is a controlled external stress test relative to the previous internal DPLL benchmarks.

2 Benchmark Families

The instances are generated from standard SAT benchmark families rather than from previous repository-internal artifacts. No external CNF files are redistributed. The six families are:

- random 3-SAT near and around the classical phase-transition region,
- planted satisfiable 3-SAT,
- modular/community-structured 3-SAT,
- random 3-XOR constraints encoded into CNF,
- graph coloring CNF encodings,
- pigeonhole-principle CNF encodings.

Table 1: Generated SAT/CDCL benchmark families.

Family	Instances	Mean target	Std.	Max target	Glucose timeouts
Graph coloring	72	1.8543	0.7838	4.4749	0
Modular 3-SAT	72	2.2894	0.5101	3.3446	0
Pigeonhole	5	7.2301	2.7275	10.1349	1
Planted 3-SAT	96	2.1401	0.7405	3.7003	0
Random 3-SAT	96	2.5785	0.8653	4.1305	0
3-XOR CNF	54	2.2972	0.5788	3.0438	0

The pigeonhole family is intentionally small but high-stress. It provides a classical unsatisfiable combinatorial obstruction rather than a random phase-transition sample.

3 CDCL Hardness Target

Each instance is solved with Glucose3 and MiniSat22 through PySAT. For each solver run we collect:

$$C = \text{conflicts}, \quad D = \text{decisions}, \quad P = \text{propagations}. \quad (1)$$

The benchmark target is

$$y_{\text{CDCL}} = \log(1 + \overline{C} + 0.10 \overline{D} + 0.01 \overline{P}). \quad (2)$$

The overbar denotes the mean over Glucose3 and MiniSat22. A conflict budget of 20000 is used. Only one instance reaches the budget in each solver, and both solvers agree on the satisfiability status for all instances after including the common budget convention.

This target is still a proxy. It is more realistic than the DPLL target of Papers 50–50b, but it is not a complete solver-independent hardness definition.

4 Feature Sets

The comparison uses six feature sets.

Feature set	Description
Density-only	Variables, clauses, clause density, and a phase-peak proxy.
Standard graph	Degree dispersion, literal imbalance, variable graph degree, clustering, components, expansion, and backdoor-density proxies.
Scalar structural	Compressed supports (H, B, S, V) , R_{rob} , and scalar cost F_{scalar} .
Graph-local structural	Scalar supports plus unit-propagation, contradiction, covariance-conditioning, expansion, and backdoor proxies.
Multi-scale defect field	Mean, tail, cluster, variance, and scale components of the local frustration field.
Shuffled defect null	The multi-scale defect-field columns independently shuffled across instances.

The multi-scale representation follows the lesson of Papers 50b–50c: do not collapse SAT structure too aggressively into one scalar. Instead, retain mean, tail, hotspot, cluster, and scale-sensitive components:

$$F_{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}}. \quad (3)$$

5 Evaluation Protocol

The main evaluation uses five-fold cross-validation with fixed random seed. Two regressors are tested:

- ridge regression after standardization,
- random forest regression.

The reported metrics are

$$R^2, \quad \text{RMSE}, \quad \rho_{\text{S}}. \quad (4)$$

A stricter leave-family-out test is also included. That test trains on five benchmark families and predicts the held-out sixth family. It is intentionally adversarial because it asks whether a feature language generalizes across qualitatively different SAT encodings.

6 Results

6.1 Main Model Comparison

Table 2: Five-fold random-forest prediction of CDCL hardness.

Feature set	R^2	RMSE	Spearman ρ
Standard graph	0.2742	0.8340	0.4254
Multi-scale defect field	0.2621	0.8409	0.3984
Graph-local structural	0.2573	0.8437	0.4170
Scalar structural	0.1062	0.9255	0.3400
Shuffled defect null	-0.0609	1.0082	-0.0070
Density-only	-0.0894	1.0218	0.2503

The main result is mixed but informative. The multi-scale defect-field representation strongly beats the shuffled-defect null and scalar compression, but it does not beat the standard graph baseline. The graph-local structural representation has slightly lower R^2 than the multi-scale representation, but slightly higher rank correlation.

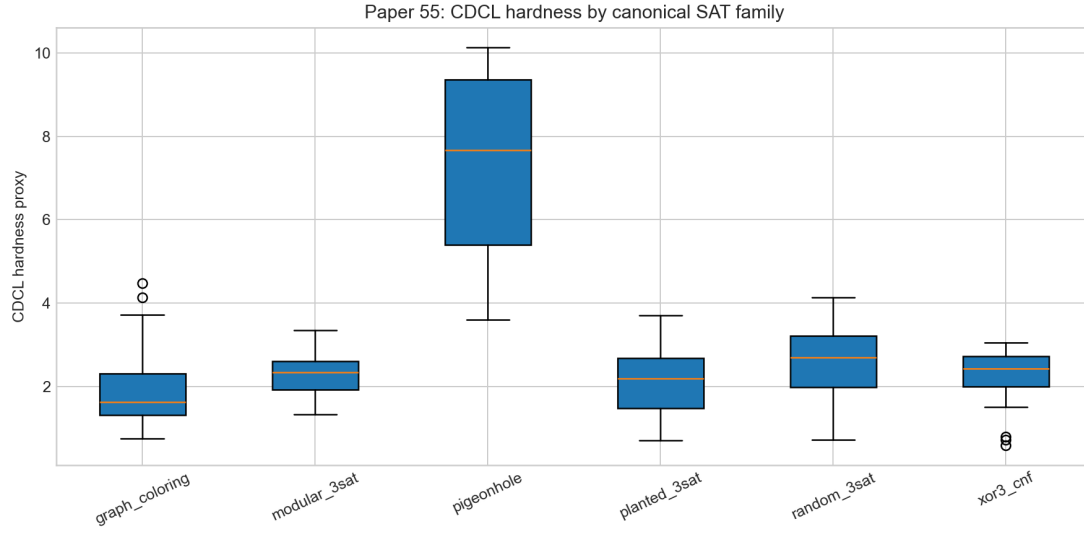


Figure 1: CDCL hardness target by canonical SAT family.

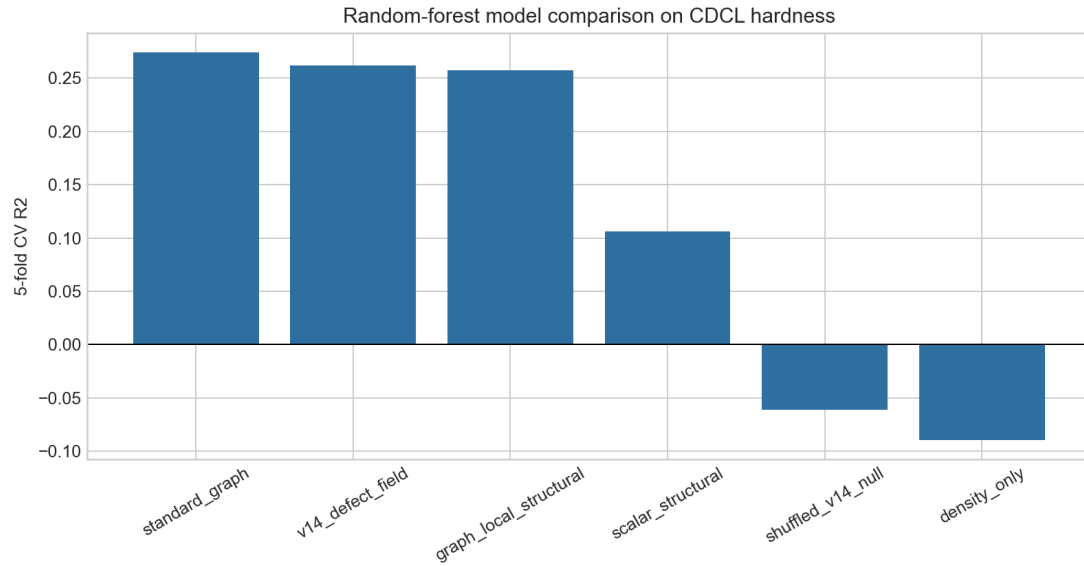


Figure 2: Random-forest model comparison. The standard graph baseline remains strongest, while multi-scale defect-field and graph-local structural features remain close and clearly above shuffled nulls.

6.2 Prediction Scatter

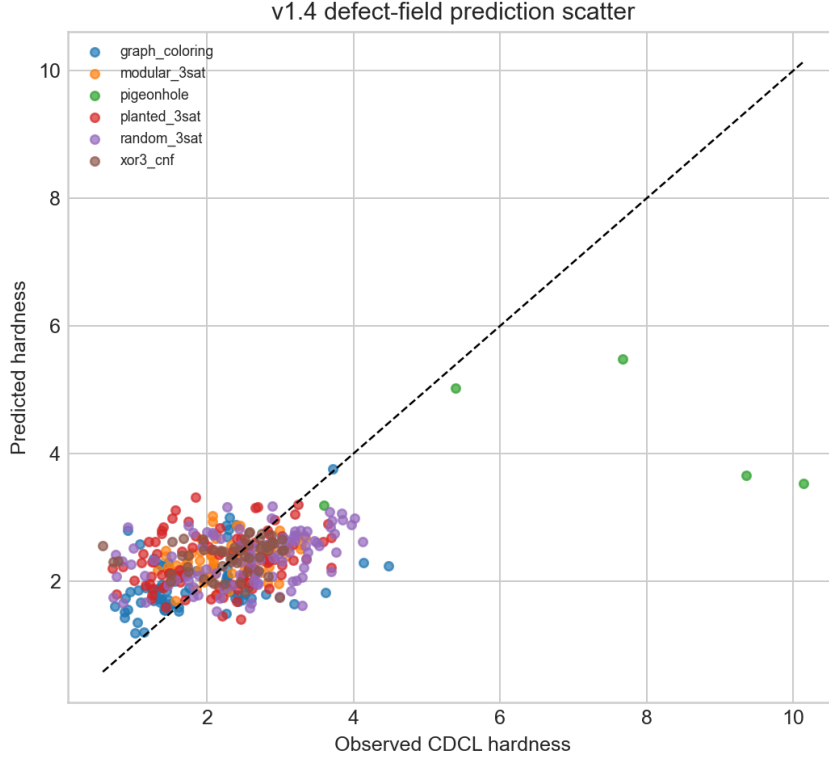


Figure 3: Out-of-fold multi-scale defect-field predictions against observed CDCL hardness.

The scatter plot shows a real but broad signal. The multi-scale feature field tracks part of the hardness variation, but family-level structure and solver-specific effects remain unresolved.

6.3 Feature Importance

Table 3: Top multi-scale defect-field permutation importances.

Feature	Importance	Std.
Frustration tail mass	0.1798	0.0162
H	0.1046	0.0083
R_{rob}	0.0690	0.0051
F_{scalar}	0.0575	0.0034
Frustration mean	0.0537	0.0034
Frustration q_{75}	0.0509	0.0047
F_{mean}	0.0439	0.0025
S	0.0413	0.0033

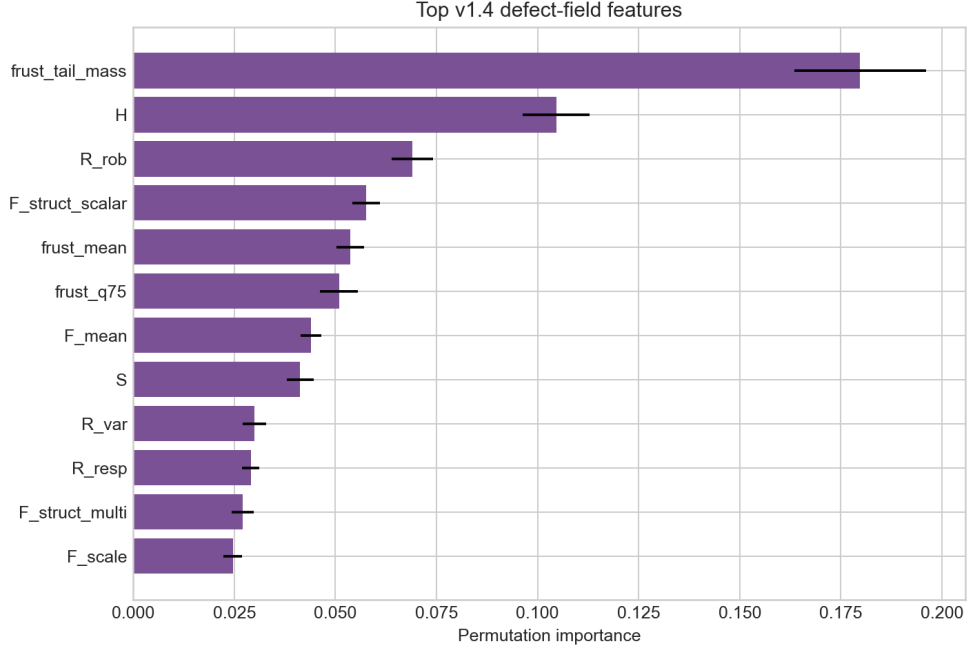


Figure 4: Top multi-scale defect-field permutation importances. Tail mass is the dominant feature, supporting the claim that rare local frustration contributes to CDCL hardness.

This is the strongest positive result for the multi-scale interpretation. The leading feature is not the mean scalar cost but the tail mass of local frustration. That supports the Paper-50b thesis that SAT hardness is sensitive to rare local structure rather than only global compression.

6.4 Leave-Family-Out Stress Test

Table 4: Overall leave-family-out performance.

Feature set	R^2	RMSE	Spearman ρ
Multi-scale defect field	-0.3104	1.1206	0.0109
Scalar structural	-0.3872	1.1530	0.0299
Density-only	-0.5228	1.2080	0.0171
Graph-local structural	-1.1692	1.4418	-0.0373
Standard graph	-2.3547	1.7930	-0.0171



Figure 5: Leave-family-out stress test. All feature sets fail to extrapolate cleanly across canonical SAT families.

This negative result is important. It is the central limitation of the benchmark: the feature sets learn within-family and mixed-family regularities, but they do not form a family-independent SAT-hardness predictor. Within mixed-family cross-validation, structural defect fields retain signal. Across held-out families, that signal does not transfer reliably.

7 Interpretation

Paper 55 gives neither a triumph nor a failure. It gives a sharper boundary.

The positive side is:

- multi-scale defect fields strongly beat shuffled-defect nulls,
- graph-local structural features are close to a standard graph baseline,
- scalar compression is clearly weaker than local defect fields,
- frustration-tail mass is the leading multi-scale feature.

The negative side is:

- standard graph features remain slightly stronger,
- leave-family-out transfer fails, which is the strongest negative result,
- the current feature language is not solver-independent,
- the benchmark is not yet based on SAT Competition or industrial CNF instances.

The most conservative conclusion is:

CDCL hardness contains signal visible to local defect-field features, but the current feature language remains close to classical graph geometry and does not yet define an independent hardness theory.

8 Limitations

- The benchmark uses generated canonical families, not downloaded SAT Competition instances.
- Only Glucose3 and MiniSat22 are used.
- Solver statistics are hardware- and implementation-dependent.
- The conflict budget censors one high-stress instance per solver.
- The feature definitions are heuristic and not derived from proof complexity.
- Leave-family-out failure shows that cross-family generalization remains unsolved.
- The benchmark does not address industrial SAT, cryptographic SAT, or large proof-complexity lower bounds.

9 Reproducibility

Repository folder:

`experiments/sat_cdcl_external_paper55/`

Main script:

```
python3 experiments/sat_cdcl_external_paper55/  
sat_cdcl_external_paper55.py
```

The script writes:

- generated instance-level solver and feature table,
- cross-validation model comparison,
- leave-family-out stress test,
- permutation importances,
- predictions,
- summary JSON,
- figures.

10 Data and License Note

No external CNF dataset is redistributed. The script generates instances from standard SAT benchmark families and solves them locally through PySAT-compatible solvers. Generated CSV, JSON, and PNG files are derived reproducibility artifacts. PySAT, Glucose, and MiniSat should be cited when reusing solver-based results. No endorsement by the solver authors or maintainers is implied.

11 Conclusion

Paper 55 replaces the earlier DPLL hardness target with CDCL solver statistics on standard SAT benchmark families. This makes the SAT track more credible and also more demanding.

The outcome is restrained. The proposed multi-scale defect-field features carry real signal and beat shuffled nulls, but they do not beat a strong standard graph baseline. The failure of leave-family-out transfer is especially important: SAT hardness is not captured by a universal scalar or by the present cross-family feature language.

The paper therefore strengthens the local-defect-field direction while weakening any broad universality claim. The next necessary step is a genuine SAT-Competition or crafted-instance benchmark with modern CDCL solvers and solver-trace targets such as conflicts, learned clauses, restarts, and proof size.

References

- [1] N. Eén and N. Sörensson, “An Extensible SAT-solver,” in *Theory and Applications of Satisfiability Testing*, SAT 2003.
- [2] G. Audemard and L. Simon, “Predicting Learnt Clauses Quality in Modern SAT Solvers,” in *Proceedings of IJCAI* (2009).
- [3] A. Ignatiev, A. Morgado, and J. Marques-Silva, “PySAT: A Python Toolkit for Prototyping with SAT Oracles,” in *Theory and Applications of Satisfiability Testing*, SAT 2018.
- [4] S. A. Cook, “The Complexity of Theorem-Proving Procedures,” in *Proceedings of the Third Annual ACM Symposium on Theory of Computing* (1971).
- [5] M. Mézard, G. Parisi, and R. Zecchina, “Analytic and Algorithmic Solution of Random Satisfiability Problems,” *Science* **297**, 812–815 (2002).